YEAR 12 · INTRODUCTION TO LOGARITHMS · WORKSHEET

Counting Multiplications

scale, bases, three kinds of graph paper, and the laws of logarithms

Name _____ Date _____

A logarithm counts multiplications: $\log_b N$ asks how many factors of b turn 1 into N, so

$$b^x = N \quad \Longleftrightarrow \quad \log_b N = x.$$

Because a logarithm is an exponent, each index law reads directly as a law of logarithms, and a graph whose axis is spaced by multiplications places every number by its logarithm.

PART A

A sense of scale

1. A line 1 km long runs from 0 to 1 000 000 000. How far from zero, in sensible units, would you mark

   (a) 100 000 _____ (b) 10 000 000 _____ (c) 1 000 _____

2. Write each as a single power of ten, and say what happens to the count of zeros.

   (a) $10^3 \times 10^4$ (b) $10^7 \div 10^2$ (c) $\left(10^2\right)^3$ (d) $10^5 \div 10^5$

3. Saving £10 000 a year, how many years does it take to save £1 000 000 000? A fingernail grows about 4 cm a year; how long would it take to grow 1 km? Use powers of ten throughout.

4. On a line spaced by $\times 10$ steps, which number sits exactly halfway between 1 and 10 000? On a line spaced by doublings, which number sits halfway between 2 and 32? Explain using exponents.

PART B

Bases and the logarithm

5. Fill in the missing exponents.

$$2^{\square} = 64 \qquad 3^{\square} = 81 \qquad 10^{\square} = 0.01 \qquad 5^{\square} = 1 \qquad 4^{\square} = 2$$

6. Complete the table, translating each statement into the other form.

Power form	Logarithm form
$2^5 = 32$	$\log_2 32 = 5$
$3^4 = 81$	
	$\log_{10} 1000 = 3$
$5^{-2} = \frac{1}{25}$	
	$\log_4 2 = \frac{1}{2}$
$b^x = N$	

7. Evaluate without a calculator, checking each answer with a power.

(a) $\log_2 32$ (b) $\log_3 \frac{1}{9}$ (c) $\log_{10} 0.001$ (d) $\log_7 1$ (e) $\log_9 3$

8. Complete the place-value table to write 19.375 in base 2.

Place	2^4	2^3	2^2	2^1	2^0	2^{-1}	2^{-2}	2^{-3}
Value	16	8	4	2	1	$\frac{1}{2}$	$\frac{1}{4}$	$\frac{1}{8}$
Digit								

$19.375 = 1 \times 10^1 + 9 \times 10^0 + 3 \times 10^{-1} + 7 \times 10^{-2} + 5 \times 10^{-3} =$ _____$_2$

Then convert 101.11_2 and 1101.1_2 to base ten.

9. Between which two consecutive whole numbers does each logarithm lie? Use your answers to say how many digits each number needs in the given base.

(a) $\log_2 50$ (b) $\log_{10} 2026$ (c) $\log_2 1000$

PART C

Three kinds of graph paper

Complete the table of values; give non-integer values to three significant figures. Then plot every relationship on each of the three graph papers that follow, joining the points with a smooth curve or a ruled line, and labelling each.

x	1	2	3	4	5	6	8	10
(i) $y = 3 + 2x$	5	7						
(ii) $y = 2^x$	2	4						
(iii) $y = x^2$	1	4						
(iv) $y = x^3$	1	8						
(v) $y = 100/x$	100	50						
(vi) $y = x^2 + 10x$	11	24						

10. Before plotting, predict which relationships will be straight on each paper. After plotting, record what you actually see.

	linear / linear	linear x / log y	log x / log y
(i) arithmetic			
(ii) geometric			
(iii) square			
(iv) cube			
(v) inverse			
(vi) two-term polynomial			

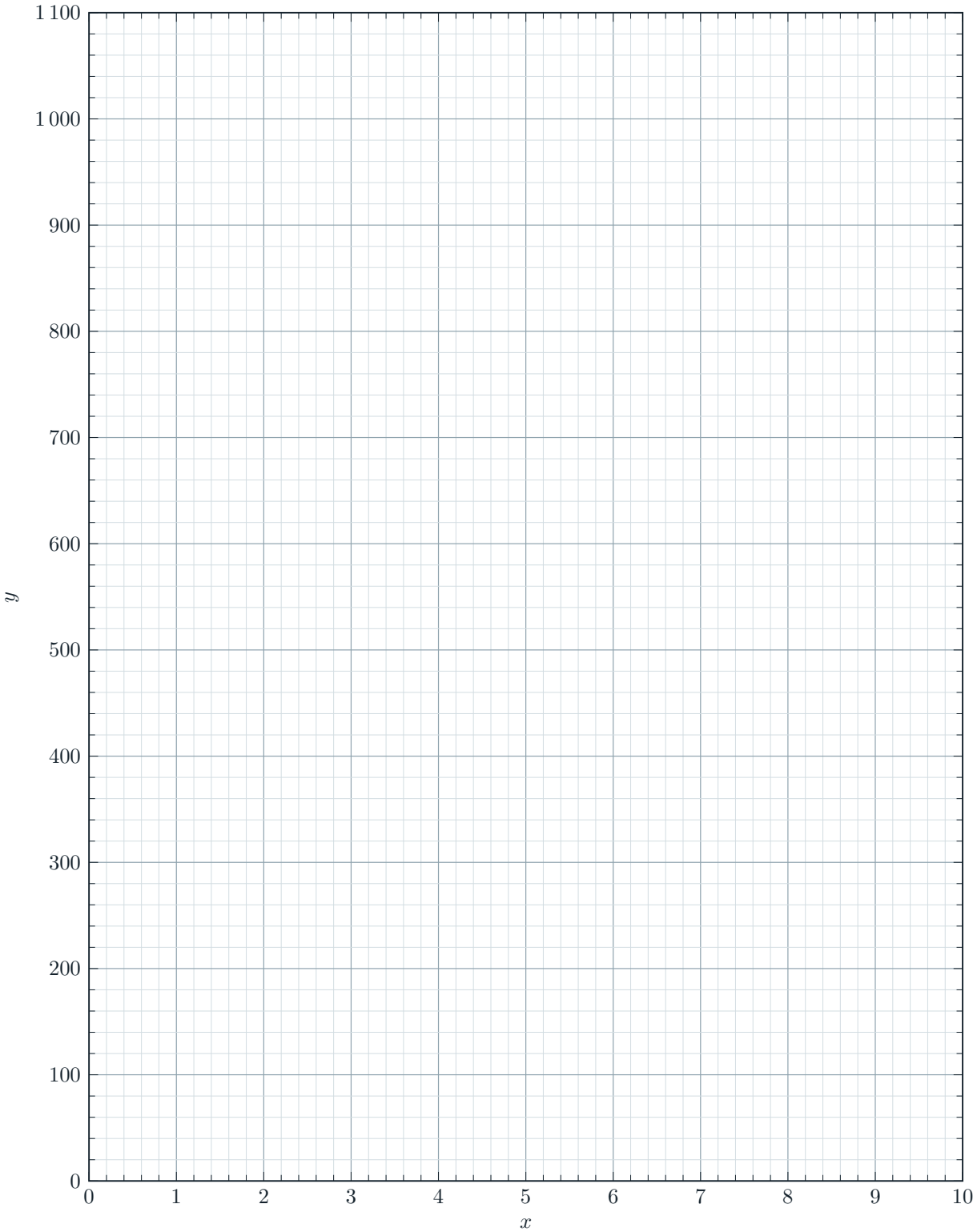
11. On the log-log paper, find the gradient of each straight line by using $\dfrac{\log_{10} y_2 - \log_{10} y_1}{\log_{10} x_2 - \log_{10} x_1}$, reading two points from your own line. What do you notice about the gradients of (iii), (iv) and (v)?

12. On the log-log paper, describe the shape of (vi). Which straight line does it resemble when x is small, and which does it approach as x grows? At what value of x are its two terms equal?

13. On the semi-log paper, by what factor does $y = 2^x$ grow each time x increases by 1, and how does the paper show that factor?

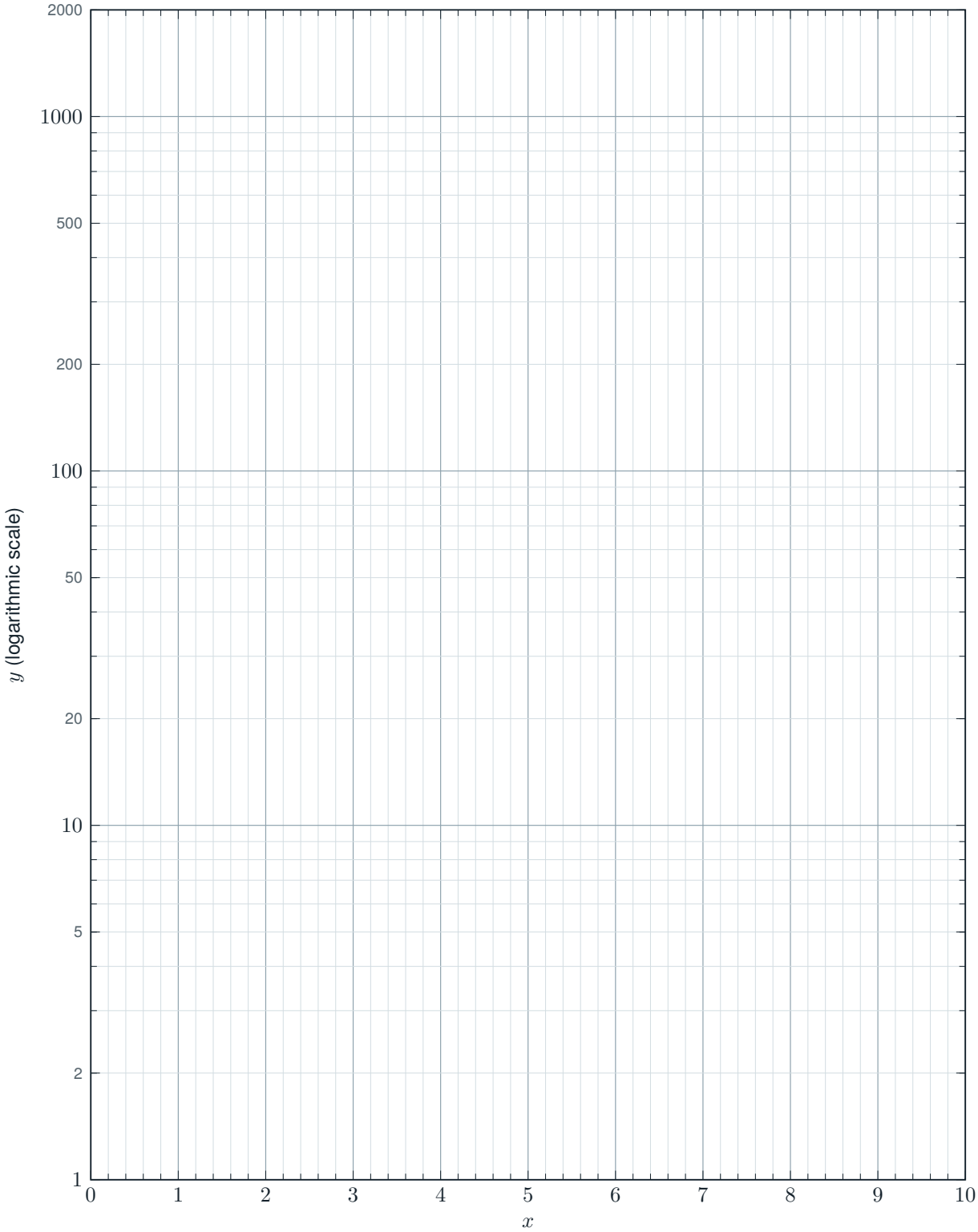
GRAPH PAPER 1 · LINEAR x / LINEAR y

Equal distances mean equal differences on both axes.



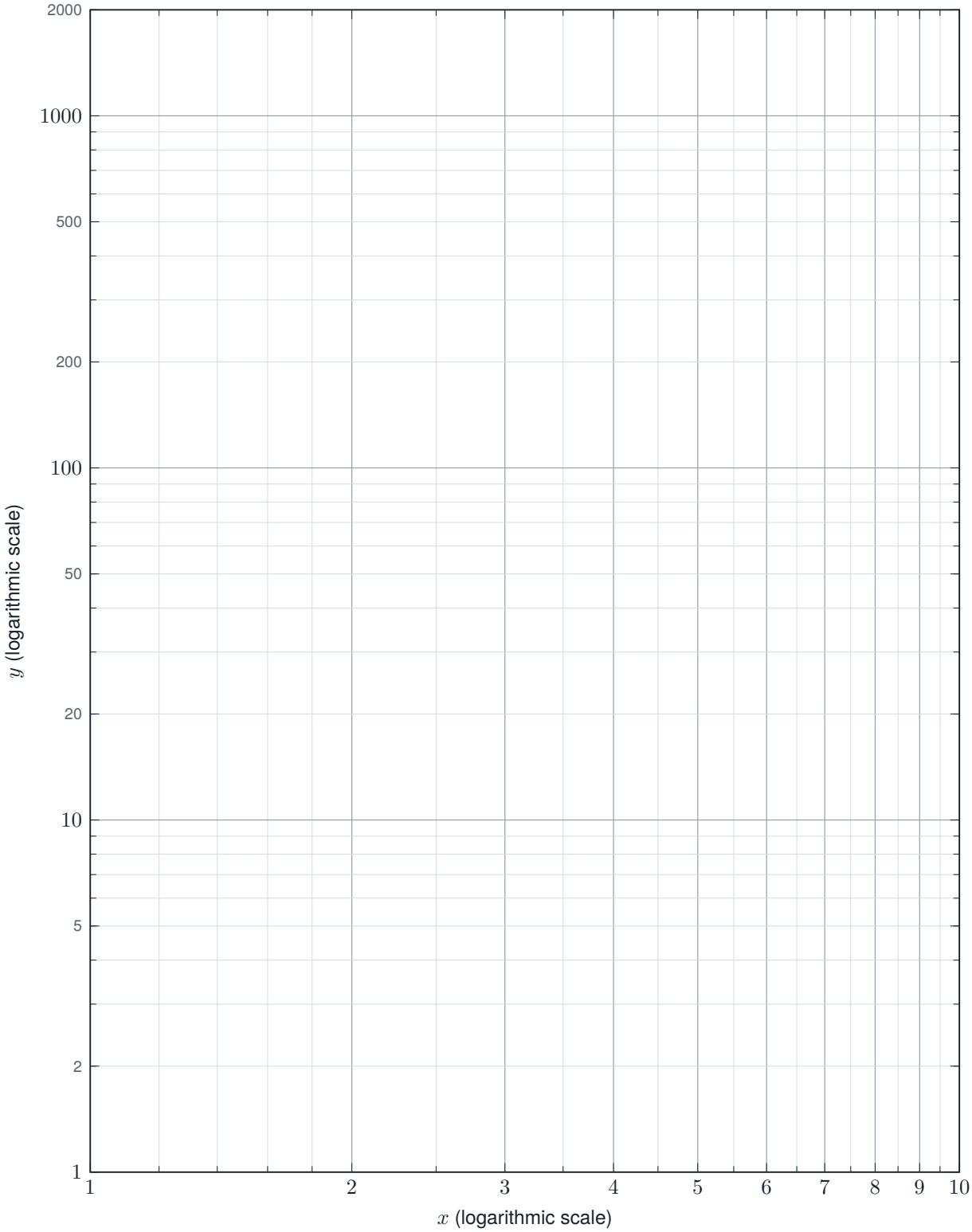
GRAPH PAPER 2 · LINEAR x / LOGARITHMIC y (SEMI-LOG)

Equal distances across mean equal differences; equal distances up mean equal multiplications.



GRAPH PAPER 3 · LOGARITHMIC x / LOGARITHMIC y (LOG-LOG)

Equal distances on either axis mean equal multiplications; plot (iii) to (vi), and (ii) for as far as it fits.



PART D

From index laws to log laws

	index law	log law
multiply	$b^m \times b^n = b^{m+n}$	$\log_b(MN) = \log_b M + \log_b N$
divide	$b^m \div b^n = b^{m-n}$	$\log_b(M/N) = \log_b M - \log_b N$
power	$(b^m)^k = b^{mk}$	$\log_b(M^k) = k \log_b M$
zero	$b^0 = 1$	$\log_b 1 = 0$

14. Write each as a single logarithm, and evaluate where possible.

(a) $\log_2 8 + \log_2 4$ (b) $\log_3 54 - \log_3 2$ (c) $2\log_5 10 - \log_5 4$ (d) $2\log x - \log y$

15. Expand fully. (a) $\log(x^2 y)$ (b) $\log \dfrac{x}{y^3}$ (c) $\log \sqrt{x}$

16. Evaluate without a calculator.

(a) $\log_6 4 + \log_6 9$ (b) $\log_2 40 - \log_2 5$ (c) $\frac{1}{2}\log_{10} 10\,000 - \log_{10} 0.1$

17. True or false? Correct each false statement.

(a) $\log(a+b) = \log a + \log b$

(b) $\log(ab) = \log a \times \log b$

(c) $\log(a^k) = k \log a$

(d) $\log_b 1 = 0$ for every base b

(e) $\log \dfrac{a}{b} = \dfrac{\log a}{\log b}$

(f) $\log_2(4+4) = \log_2 4 + \log_2 4$

18. Take logarithms of each relationship and write it in the form $Y = mX + c$. State which graph paper makes it straight, and give the gradient and intercept.

(a) $y = 4x^3$ (b) $y = 5 \times 3^x$ (c) $y = 200\,x^{-1/2}$

19. Working backwards from a straight line.

(a) On log-log paper a straight line passes through $(1, 2)$ and $(10, 200)$. Find k and n in $y = kx^n$.

(b) On semi-log paper a straight line passes through $(0, 3)$ and $(4, 48)$. Find a and b in $y = a\,b^x$.

20. Changing base. Since $10 \approx 2^{3.32}$, every $\times 10$ step is worth about 3.32 doublings.

    (a) Given $\log_{10} 2 \approx 0.301$, show that $\log_2 1000 \approx 9.97$, and hence say how many binary digits 1000 needs.

    (b) Given $\log_{10} 19.375 \approx 1.287$, estimate $\log_2 19.375$ and check it against your answer to question 8.

21. In one or two sentences of your own: what does $\log_b N$ count, and why does that make multiplication turn into addition?

FOR THE TEACHER

Answers

1. The line has 10^{-6} m per unit, so (a) 10 cm, (b) 10 m, (c) 1 mm.

2. (a) 10^7, counts add; (b) 10^5, counts subtract; (c) 10^6, counts multiply; (d) $10^0 = 1$.

3. Saving takes $10^9 \div 10^4 = 10^5 = 100\,000$ years. The nail needs 1 km $= 10^5$ cm at about 4 cm a year, roughly $25\,000$ years, so the nail wins by a factor of about four.

4. 100, since the exponents 0 and 4 average to 2; and 8, since $2 = 2^1$ and $32 = 2^5$ average to 2^3. Each is the geometric mean, $\sqrt{1 \times 10\,000}$ and $\sqrt{2 \times 32}$.

5. 6, 4, -2, 0, $\frac{1}{2}$.

6. $\log_3 81 = 4$; $10^3 = 1000$; $\log_5 \frac{1}{25} = -2$; $4^{1/2} = 2$; $\log_b N = x$.

7. (a) 5 as $2^5 = 32$; (b) -2 as $3^{-2} = \frac{1}{9}$; (c) -3; (d) 0; (e) $\frac{1}{2}$ as $9^{1/2} = 3$.

8. Digits 1 0 0 1 1 . 0 1 1, so $19.375 = 10011.011_2$, which stops because $0.375 = \frac{3}{8} = \frac{1}{4} + \frac{1}{8}$. Also $101.11_2 = 4 + 1 + \frac{1}{2} + \frac{1}{4} = 5.75$ and $1101.1_2 = 8 + 4 + 1 + \frac{1}{2} = 13.5$.

9. (a) $5 < \log_2 50 < 6$, as $32 < 50 < 64$, so 50 needs 6 binary digits (110010_2); (b) $3 < \log_{10} 2026 < 4$, so 4 decimal digits; (c) $9 < \log_2 1000 < 10$, as $512 < 1000 < 1024$, so 10 binary digits.

Table of values.

x	1	2	3	4	5	6	8	10
$3 + 2x$	5	7	9	11	13	15	19	23
2^x	2	4	8	16	32	64	256	1024
x^2	1	4	9	16	25	36	64	100
x^3	1	8	27	64	125	216	512	1000
$100/x$	100	50	33.3	25	20	16.7	12.5	10
$x^2 + 10x$	11	24	39	56	75	96	144	200

10. (i) is straight only on linear paper; (ii) only on semi-log paper; (iii), (iv) and (v) only on log-log paper; (vi) is exactly straight on none of them, although on log-log paper it is close to straight over short stretches.

11. Gradients 2, 3 and -1: on log-log paper the gradient of $y = kx^n$ is the power n.

12. For small x the $10x$ term dominates, so the curve runs close to a line of gradient 1; for large x the x^2 term dominates and it approaches a line of gradient 2. The terms are equal when $x^2 = 10x$, at $x = 10$, which is where the bend is most pronounced. Only single-term power laws are exactly straight on log-log paper.

13. y doubles for every $+1$ in x, so each unit across rises by the same distance up the paper, $\log_{10} 2 \approx 0.301$ of a decade.

14. (a) $\log_2 32 = 5$; (b) $\log_3 27 = 3$; (c) $\log_5 \dfrac{10^2}{4} = \log_5 25 = 2$; (d) $\log \dfrac{x^2}{y}$.

15. (a) $2\log x + \log y$; (b) $\log x - 3\log y$; (c) $\frac{1}{2}\log x$.

16. (a) $\log_6 36 = 2$; (b) $\log_2 8 = 3$; (c) $\frac{1}{2} \times 4 - (-1) = 3$, or as one logarithm $\log_{10} \dfrac{\sqrt{10\,000}}{0.1} = \log_{10} 1000 = 3$.

17. (a) false, there is no law for the logarithm of a sum; (b) false, $\log(ab) = \log a + \log b$; (c) true; (d) true, since $b^0 = 1$; (e) false, $\log \frac{a}{b} = \log a - \log b$; (f) false, $\log_2 8 = 3$ whereas the right-hand side is 4.

18. (a) $\log y = 3\log x + \log 4$: log-log, gradient 3, intercept $\log 4 \approx 0.602$. (b) $\log y = (\log 3)\,x + \log 5$: semi-log, gradient $\log 3 \approx 0.477$, intercept $\log 5 \approx 0.699$; in base 3 this is $\log_3 y = x + \log_3 5$. (c) $\log y = -\frac{1}{2}\log x + \log 200$: log-log, gradient $-\frac{1}{2}$, intercept $\log 200 \approx 2.301$.

19. (a) $n = \dfrac{\log 200 - \log 2}{\log 10 - \log 1} = 2$ and $k = 2$, so $y = 2x^2$. (b) $a = 3$ and $b^4 = 16$, so $b = 2$ and $y = 3 \times 2^x$.

20. (a) $\log_2 1000 = \dfrac{\log_{10} 1000}{\log_{10} 2} \approx \dfrac{3}{0.301} \approx 9.97$, so 1000 lies between 2^9 and 2^{10} and needs 10 binary digits. (b) $\log_2 19.375 \approx 1.287 \div 0.301 \approx 4.28$, between 4 and 5, matching the five binary digits of 10011.011_2 before the point.

21. $\log_b N$ counts how many factors of b make N; multiplying two numbers joins their lists of factors, so the counts add.